

4(a)	any two from: cellulose molecules 1 straight (chains) / linear / unbranched ; 2 arranged in parallel ; 3 hydrogen bonds between (cellulose) molecules ; 4 any detail, e.g. between H of –OH groups and –O of –OH groups ; A between -OH groups 5 AVP ; e.g. cellulose molecules have, staggered / overlapping, ends I overlapping microfibrils	2
4(b)(i)	any three from: 1 two layers of phospholipid ; 2 label for, phospholipids / phospholipid bilayer ; 3 channel protein across whole bilayer with <u>hole through middle</u> + label ; ecf if a monolayer shown or no phospholipids drawn 4 any additional feature + label ; e.g. hydrophilic heads / hydrophobic tails / cholesterol / other named protein (e.g. glycoprotein)	3

Question	Answer	Marks
4(b)(ii)	any two from: 1 to allow <u>facilitated diffusion</u> ; R active transport 2 ions are, charged / hydrophilic ; 3 cannot pass through, hydrophobic core / phospholipid bilayer ; A ora – channel proteins provide a, hydrophilic / water-filled, pathway (through the membrane) / AW A channel proteins are lined by polar R groups A ions are repelled by, hydrophobic core / phospholipid bilayer	2
4(c)	any three from: 1 transpiration (rate), will not decrease to a low level / remains the same (if conditions remain the same) / remain the same at night / increases if (named) condition(s) change / AW ; 2 <u>water vapour</u> continues to <u>diffuse out</u> (from intercellular air spaces) ; 3 (because) stomata, cannot close / remain open ; 4 guard cells do not lose turgidity ; 5 because no outward flow of water (from guard cells) ; 6 by osmosis / down water potential gradient (from guard cells) ; 7 AVP ; e.g. <i>ref. to</i> more likely to wilt rates of gaseous exchange remain the same	3